

Chapter 1

Overview Of Operating System

1.1. OS- Concept, Components of OS, Operations of OS, Views of OS

A computer system has many resources (hardware and software), which may be required to complete a task. The commonly required resources are input/output devices, memory, file storage space, CPU etc. The operating system acts as a manager of the above resources and allocates them to specific programs and users as necessary for their task. Therefore operating system is the resource manager i.e. it can manage the resource of a computer system internally. The resources are processor, memory, files, and I/O devices.

An OS is a program that manages the computer hardware. It also provides a basis for application programs and acts as an intermediary between the computer user and the computer hardware.

Operating System (OS) is an interface between computer user and computer hardware

OR

An operating system is software which performs all the basic tasks like file management, memory management, process management, handling input and output, and controlling peripheral devices such as disk drives and printers.

Example Operating Systems :- Linux, Windows, OS X, VMS, OS/400, AIX, z/OS, etc.

OR

An operating system is a program that acts as an interface between the user and the computer hardware and controls the execution of all kinds of programs.

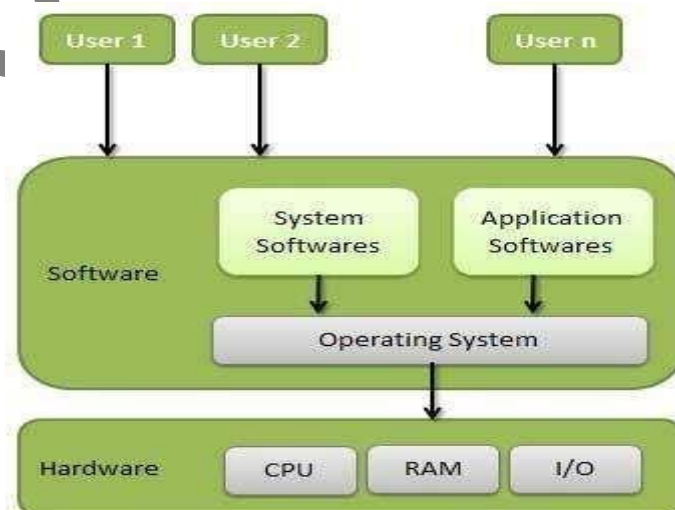


Fig.1: Operating System

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OS acts as a control program which controls all activities of computer system. It interacts with hardware of computer system to perform different tasks.

Primary objective of OS is to make computer convenient to use and utilize hardware in efficient manner.

Computer system can be divided into 4 components:

1. Hardware
2. Operating System
3. Application Program
4. Users

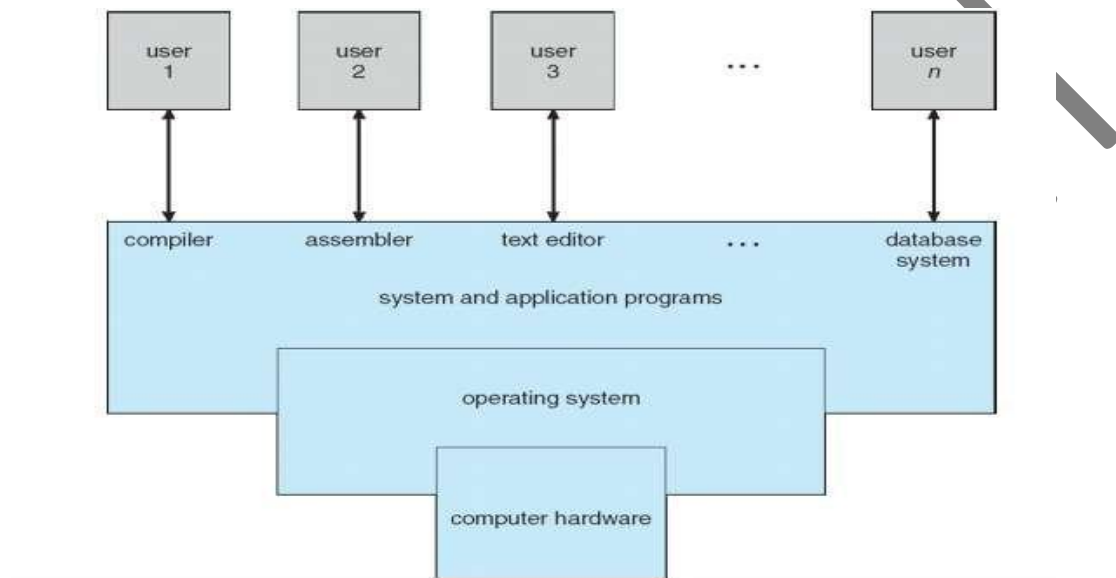


Fig.2 : Components of Computer System

1. **Hardware:** These are physical parts of machine which provides basic computing resources. It includes CPU, memory, I/O devices etc. which provides basic computing resources for system.
2. **Application Program:** It defines the way in which these resources are used to solve user computing problems. e.g. word processor, compiler, browser
3. **Operating System:** It controls & coordinate the use of hardware amongst the various application program for various users.
4. **Users:** Users are human beings, machines and other computers.

An operating system is similar to a government. Like a government, it performs no useful function by itself. It simply provides an environment within which other programs can do useful work. To understand more fully the operating systems role, we next explore operating systems from two view points :-from users point and from system point

Views of OS

1. User View

- The user's view of the computer varies according to the interface being used.
- Most computer users sit in front of a PC, consisting of a monitor/ keyboard/ mouse, and system unit.
- The goal is to maximize the work (or play) that the user is performing.
- In this case/ the operating system is designed mostly for ease of use with some attention paid to resource utilization-how performance and none paid to various hardware and software resources are shared. Performance is, of

course, important to the user; but such systems are optimized for the single-user experience rather than the requirements of multiple users.

- In other cases, a user sits at a terminal connected to a mainframe or minicomputer. Other users are accessing the same computer through other terminals. These users share resources and may exchange information. The operating system in such cases is designed to maximize resource utilization. - to assure that all available CPU time, memory, and I/O are used efficiently and that no individual user takes more than her fair share.
- In still other cases, users sit at workstations connected to networks of other workstations and These users have dedicated resources at their disposal, but they also share resources such as networking and servers-file, compute, and print servers. Therefore, their operating system is designed to compromise between individual usability and resource utilization.
- Recently, many varieties of handheld computers have come into fashion. Most of these devices are standalone units for individual users. Some are connected to networks, either directly by wire or (more often) through wireless modems and networking. Because of power, speed, and interface limitations, they perform relatively few remote operations. Their operating systems are designed mostly for individual usability, but performance per unit of battery life is important as well.
- Some computers have little or no user view

2. System View

- From the computer's point of view, the operating system is the program most intimately involved with the hardware.
- In this context, we can view an operating system as a resource allocator.
- A computer system has many resources that may be required to solve a problem: CPU time, memory space, file-storage space, I/O devices, and so on. The operating system acts as the manager of these resources. As we have seen, resource allocation is especially important where many users access the same mainframe or minicomputer.
- A slightly different view of an operating system emphasizes the need to control the various I/O devices and user programs. An operating system is a control program. A control program manages the execution of user programs to prevent errors and improper use of the computer. It is especially concerned with the operation and control of I/O devices.

Operations of OS

- Transferring input from input device to memory.
- Transferring input from memory to CPU.
- Transferring data from CPU to output device.
- Copy or move data from one storage device to other.
- It manages the programs running on the system
- Display appropriate error messages in case of software or hardware problems.
- Maintain details of files & directories.
- To provide security to users data.
- Protection of storage devices from overwriting.
- It manages the hardware and software resources of the system.
- It provides a stable, consistent way for applications to deal with the hardware without knowing all the details of the hardware.
- Operations of os must consider following operations

- **Program Management**
- **Resource Management**
 - **Memory Management**
 - **File System Management**
 - **I/O subsystem Management**
- **Security And Protection.**

- **Process Management**

A process is a program in execution. It is a unit of work within the system. Program is a *passive entity*; process is an *active entity*.

Process needs resources to accomplish its task

- CPU, memory, I/O, files
- Initialization data

Process termination requires reclaim of any reusable resources

Single-threaded process has one program counter specifying location of next instruction to execute

- Process executes instructions sequentially, one at a time, until completion

Multi-threaded process has one program counter per thread

Typically system has many processes, some user, some operating system running concurrently on one or more CPUs

- Concurrency by multiplexing the CPUs among the processes / threads

Process Management Activities

The operating system is responsible for the following activities in connection with process management:

- Creating and deleting both user and system processes
- Suspending and resuming processes
- Providing mechanisms for process synchronization
- Providing mechanisms for process communication
- Providing mechanisms for deadlock handling

- **Resource Management**

Memory Management

To execute a program all (or part) of the instructions must be in memory

All (or part) of the data that is needed by the program must be in memory

Memory management determines what is in memory and when

- Optimizing CPU utilization and computer response to users

Memory management activities

- Keeping track of which parts of memory are currently being used and by whom
- Deciding which processes (or parts thereof) and data to move into and out of memory
- Allocating and de allocating memory space as needed

File-system Management

OS provides uniform, logical view of information storage

- Abstracts physical properties to logical storage unit - file

File-System management

- Files usually organized into directories
- Access control on most systems to determine who can access what
- OS activities include
 - Creating and deleting files and directories
 - Primitives to manipulate files and directories

- Mapping files onto secondary storage
- Backup files onto stable (non-volatile) storage media

I/O Subsystem

One purpose of OS is to hide peculiarities of hardware devices from the user

I/O subsystem responsible for

- Memory management of I/O including buffering (storing data temporarily while it is being transferred), caching (storing parts of data in faster storage for performance), spooling (the overlapping of output of one job with input of other jobs)
- General device-driver interface
- Drivers for specific hardware devices

◦ **Protection and Security**

Protection – any mechanism for controlling access of processes or users to resources defined by the OS

Security – defense of the system against internal and external attacks

- Huge range, including denial-of-service, worms, viruses, identity theft, theft of service

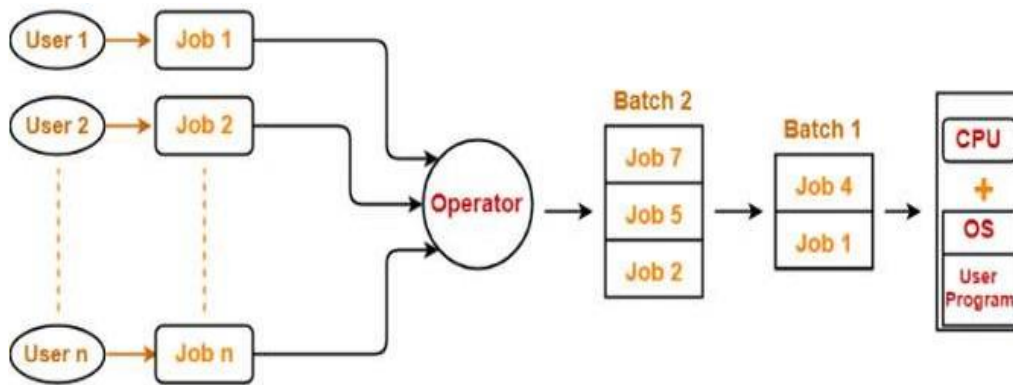
Systems generally first distinguish among users, to determine who can do what

- User identities (**user IDs**, security IDs) include name and associated number, one per user
- User ID then associated with all files, processes of that user to determine access control
- Group identifier (**group ID**) allows set of users to be defined and controls managed, then also associated with each process, file
- **Privilege escalation** allows user to change to effective ID with more rights

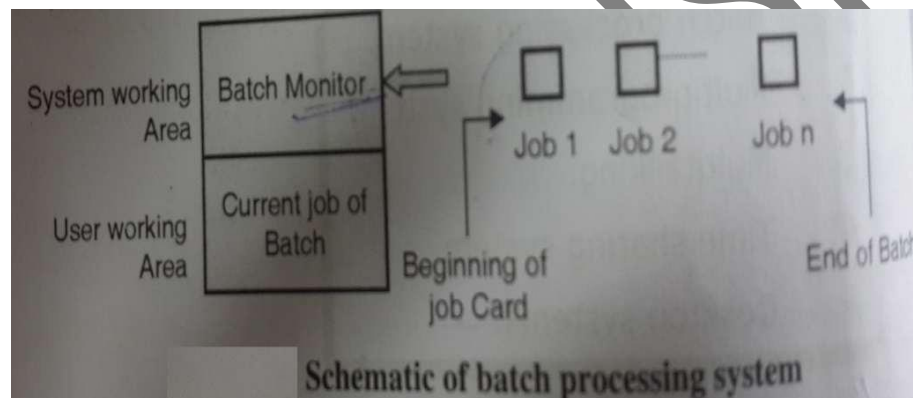
1.2 Types of Operating System

1. Batch operating System

- A batch stands for sequence of user job.
- The programmers leave their programs with the operator, operator then sorts the programs with similar requirements into batches & submit it to batch processing system (BPS) & then those jobs were executed automatically without user interaction.
- The users of a batch operating system do not interact with the computer directly.
- The jobs were recognized with the special markers which indicate start & end of jobs.
- Following fig. shows schematic of batch processing system.
- BPS consists of batch monitor or supervisor was the major component which permanently placed in part of memory & other part of memory is used to process user's job.



OR



- Batch monitor keeps control on processing environment.
- Functions of batch monitor are:
 - Scheduling:**
 - This activity determines which service request should be handled in next turn.
 - BPS uses First Come First Serve (FCFS) algorithm for scheduling.
 - Memory management:-**
 - During BPS operation memory is divided into two parts
1. System working area 2. User working area.
 - Partitioning of memory is done by OS.
 - Some part of monitor is permanently stored in memory known as resident area of monitor.
 - Sharing & protection :**
 - By dividing the memory in two parts protection is very well provided.

Advantages of Batch OS:

- Repeated jobs are done fast in batch systems without user interaction.
- You don't need special hardware and system support to input data in batch systems.
- Best for large organizations
- Sharing of batch system for multiple users.
- The idle time batch system is very less.

- You can assign specific time for the batch jobs so when the computer is idle it starts processing the batch jobs i.e. at night or any free time.
- The batch systems can manage large repeated work easily.
- Programmer attention is reduced.

Disadvantages of Batch OS:

- Monitor remains in memory it results in amount of memory wastage
- It is difficult to debug batch systems.
- Batch systems are sometime costly.
- Lack of protection

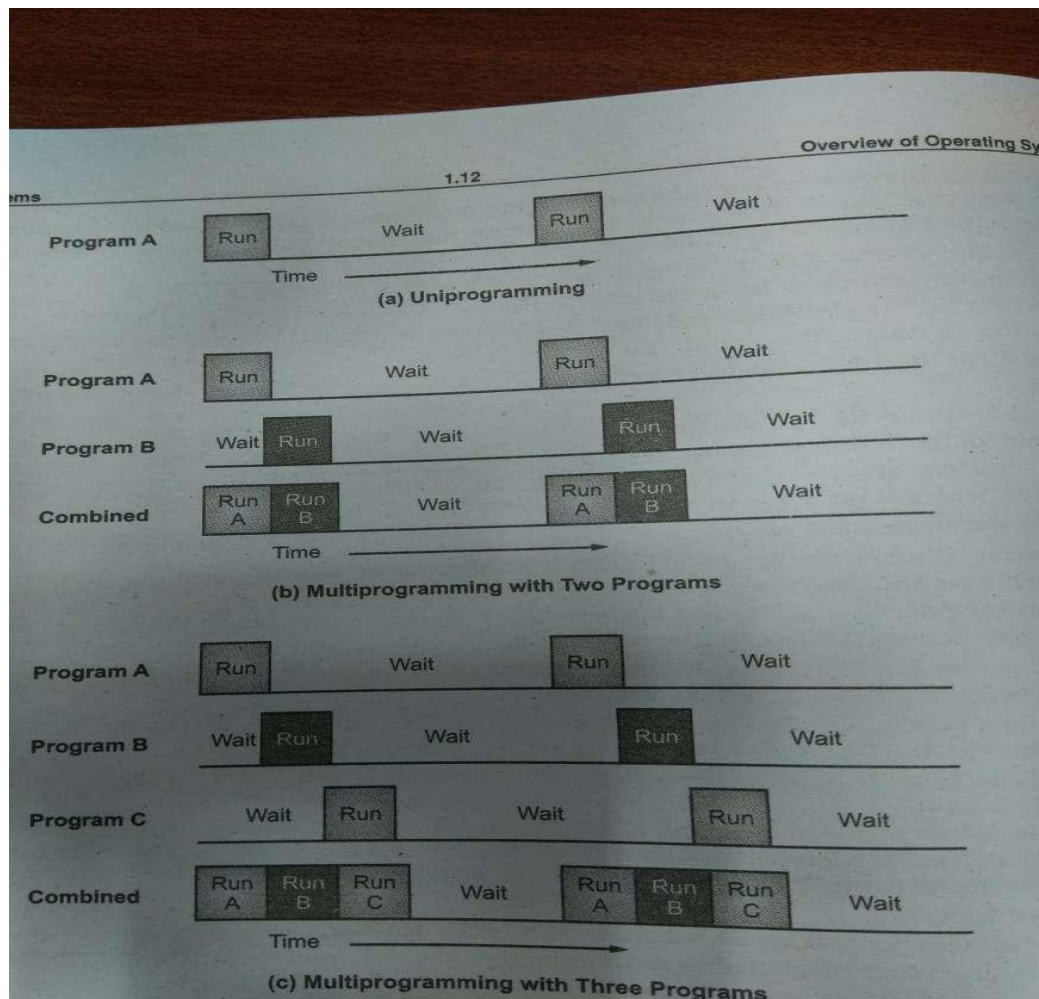
2. Multiprogramming

- The process of executing multiple programs simultaneously is known as multiprogramming.
- Uniprogramming cannot keep CPU & I/O devices busy at all time.
- Uniprogramming cannot keep I/O devices as well as CPU busy.
- Using multiprogramming we can increase utilization of CPU & I/O device.
- Multiprogramming OS keeps several jobs in memory at a time. Set of jobs are shown below



Fig. :- Memory layout of Multiprogramming.

- This set of jobs is kept in ^{job} pool.
- Following diagram shows working of uniprogramming & multiprogramming system



- By considering above diagram of multiprogramming system initially, Job A is given for execution. After some time it may require I/O operation. So the CPU becomes idle (wait). Instead of making CPU idle, it switches to next job for execution till it requires I/O operation. The CPU switches to next job and so on.
- Job may have to wait for some task. When job needs to wait, CPU switches to another job, executes it, and so on.
- E.g. A computer running excel and firefox browser simultaneously

Advantages :-

- Setup time is reduced.
- Efficiency is improved
- Waiting time is limited for programs.
- CPU utilization increases.

Disadvantages :-

- Proper memory management is required.
- Proper scheduling algorithm is required.
- Maintenance is expensive.

3. Multitasking

- Multitasking is the ability of an operating system to execute more than one task simultaneously on a single processor machine
- Multitasking is logical extension of multiprogramming.

- Though we say so but in reality no two tasks on a single processor machine can be executed at the same time.
- Actually CPU switches from one task to the next task so quickly that appears as if all the tasks are executing at the same time.
- A task is a particular operation like edit, copy, rename, delete, open, close etc.
- Multitasking is possible by executing one job in foreground & many jobs in background.
- Foreground jobs require user interaction where as background jobs does not require user interaction.
- E.g. While editing we can take printouts & listen music.



Advantages :-

- Multiple tasks can be run simultaneously
- Increases speed of execution
- CPU utilization increases.

Disadvantages:-

- These systems are not reliable.
- Problem of data communication.
- It raises question of security of user program & data.

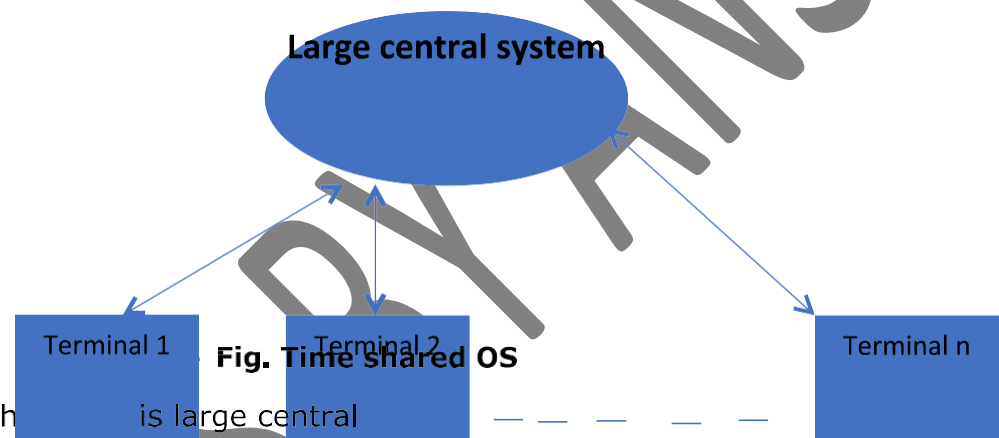
Sr. No.	Multiprogramming	Multitasking
1	Multiple programs are executed simultaneously.	Multiple tasks are executed simultaneously.
2	Collection of different tasks is program.	Task is smaller unit of program.
3	Program may contain multiple task .	Task do not contain program
4	User interaction is not provided	User interaction is provided
5	It utilizes CPU & I/O devices efficiently	It utilizes CPU efficiently
6	There are simply only programs	There are two types of task foreground & background task.

4. Time sharing OS

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4. Time sharing OS

- Time-sharing is a technique which enables program people, located at terminals, to use a particular computer system at the same time.
- Processor's time which is shared among multiple users simultaneously is termed as time-sharing.
- The main objective of Time-Sharing is to minimize response time.
- Multiple jobs are executed by the CPU by switching between them, but the switches occur so frequently. Thus, the user can receive an immediate response.
- Example OS :Linux



- In this system, a large central computer provides time slice(in milliseconds) at regular interval to each terminal. The time sliced is switched so rapidly that each terminal feels that the whole computer is dedicated to himself or herself.
- CPU does not wait for any operation for every job ,it immediately switched to to other job as the time slice is very less.

- Advantages :-
 - Provides the advantage of quick response
 - Avoids duplication of software
 - Reduces CPU idle time

- Disadvantages :-
 - Expensive to build.
 - Problem of data communication
 - Difficult & complicated.

5. Multiprocessor OS/Tightly couples/Parallel System

- In this more than one processors can be places in single cabinet which shares resources, bus, memory etc.
- **Objective** –is to get maximum work done in less time.
- It is also called as tightly coupled or parallel system.

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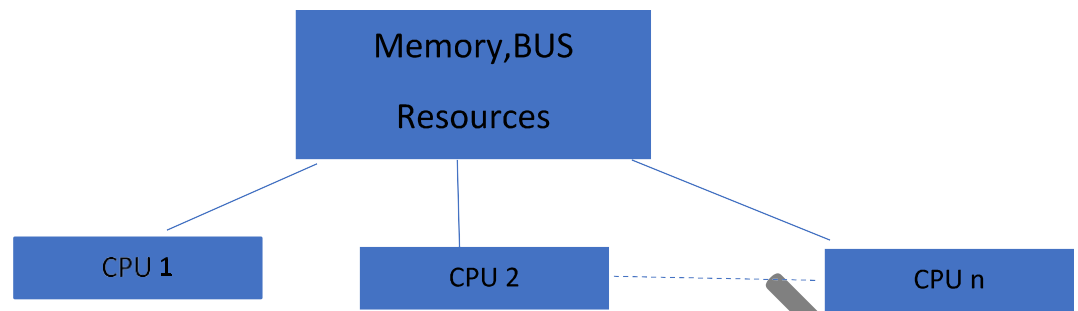


Fig. Multiprocessor OS

OR

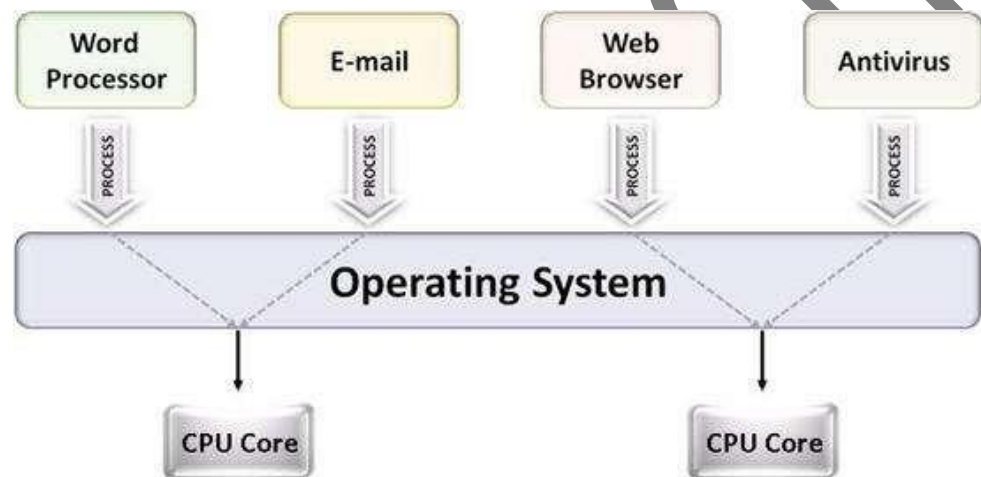


Fig. :-Multiprocessing System

- Types of Multiprocessor System are
 1. **Symmetric Multiprocessing** – In this two or more processors are connected with high bandwidth link & manage by single OS. All processor equally share I/O devices.
 2. **Asymmetric Multiprocessing** – Each processor is assigned specific task. IN this Master-Slave concept is used. Master processors controls slave processors.

Advantages:-

- It increases throughput.
- It increases reliability (if one processor fails, whole system will not fail.)
- More than one processors can be used.
- Maximum work done can be done in less time.

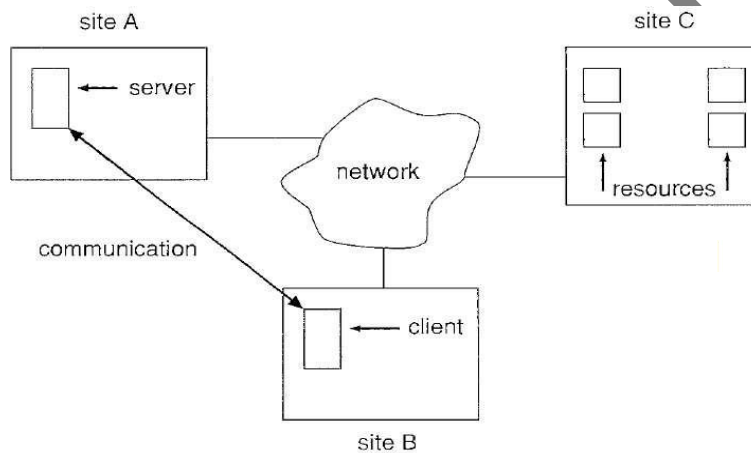
Disadvantages:-

- Expensive to build & maintain.
- More complex in both hardware & software as compare to uniprocessor system.

6. Distributed Operating System/Loosely coupled System

- It is collection of processors interconnected by communication network.

- Also called as **loosely coupled system** and are exactly opposite to multiprocessor system.
- From point of view of specific processor, rest of processor & their respective resources are remote whereas its own are local.
- Data processing jobs are distributed among the processors accordingly.
- The processors communicate with one another through various communication lines (such as high-speed buses or telephone lines).
- Processors in a distributed system may vary in size and function. These processors are referred as sites, nodes, computers, and so on.



A distributed system.

- Advantages:-**
- With resource sharing facility, a user at one site may be able to use the resources available at another.
 - Speedup the exchange of data with one another via electronic mail.
 - If one site fails in a distributed system, the remaining sites can potentially continue operating.
 - Better service to the customers.
 - Reduction of the load on the host computer.
 - Reduction of delays in data processing.

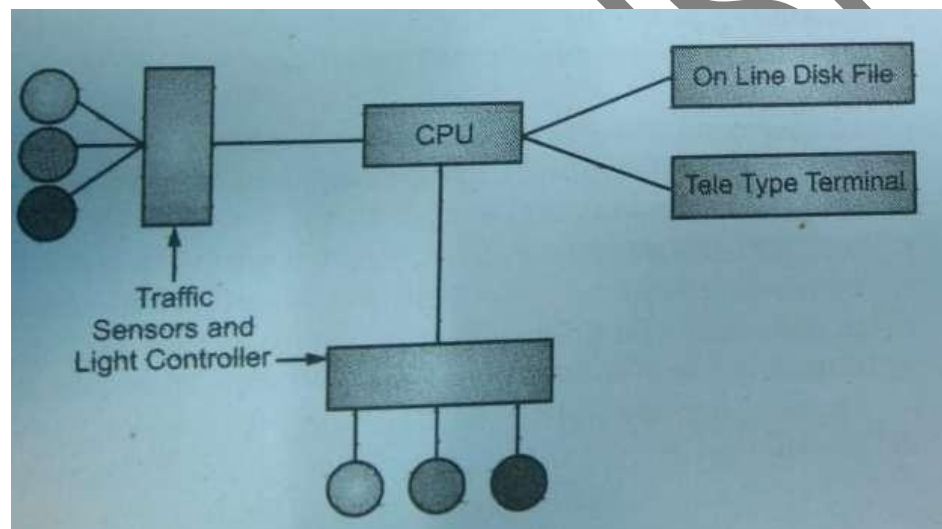
- Disadvantages:-**
- Complex & difficult to build.
 - Special security measures are needed to protect the shared resources.

7. Real time System

- Real time systems are used in environment number of events, mostly external must be accepted and processes deadline. where a large to the computer system, in short time or within
- This systems are often used application. as control device in dedicated

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- **Objective** -To provide quick event response time & meet scheduled deadline.
- It is used when rigid time requirement have been placed on operation of a process.
- **In this sensors** are used, sensors bring data to computer, computer then analyses data & adjust controls according to sensors input.
- It has fixed time constraints otherwise system will fail.
- **Examples:** Nuclear power plant control, Railway & airline reservation system, industrial control system, satellite communication, traffic light control etc.



- There are two types of real-time systems:
 - Hard real-time systems**
hard real-time systems guarantees that critical tasks complete on given time.
E.g. car engine control system.- delay in particular signal may cause engine failure.[plz illustrate example]
 - Soft real-time systems**
Soft real time systems will tolerate delay in time.
E.g. Live audio, live video [plz illustrate example]

Advantages :-

- Direct communication between user & system is possible
- System generated task can be completed within time

Disadvantages :-

- More complex & difficult to maintain

8. Mobile OS

- A mobile OS is one that controls smartphones, PDA's, tablet PC's etc.
- Like different OS such as Linux or Windows operating system controls your desktop or laptop computer, a mobile

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operating system is the software platform on top of which other programs can run on mobile devices.

- The operating system is responsible for determining the functions and features available on your device, such as thumb wheel, keyboards, WAP, synchronization with applications, email, text messaging and more.
- Different Mobile OS are Android OS (Google Inc.), Bada (Samsung Electronics) BlackBerry OS , iPhone OS / iOS (Apple), MeeGo OS (Nokia and Intel), Symbian OS (Nokia), Windows Mobile (Windows Phone)

1. Android OS

- Android is a Linux based operating system it is designed primarily for touch screen mobile devices such as smart phones and tablet computers.
- The operating system have developed a lot in last 15 years starting from black and white phones to recent smart phones or mini computers.
- One of the most widely used mobile OS these days is android.
- The android is a powerful operating system and it supports large number of applications in Smartphones. These applications are more comfortable and advanced for the users.
- The hardware that supports android software is based on ARM architecture platform. The android is an open source operating system means that it's free and any one can use it.
- The android has got millions of apps available that can help you managing your life one or other way and it is available low cost in market at that reasons android is very popular.
- **Advantages**
 - Android is Linux based open source operating system , it can be developed by any one
 - Easy access to the android apps
 - You can replace the battery and mass storage, disk drive
 - Its supports all Google services
 - It supports Multitasking
 - Its free to customize
 - Its supports 2D and 3D graphics
- **Disadvantages**
 - Usually you need more code on Java than Objective-C.
 - Complex layouts and animations are harder to code in Android.

- Applications contains virus also present in Android Market
- A lot of "process" in the background that lead to the battery quickly drains.
- Low security and fake apps can be installed to steal your info from unknown resources

2. iOS

- This OS is developed by Apple.
- It is OS that presently powers many of the companies mobile devices including iPhone, iPad and iPod.
- It is the second most popular OS globally after Android
- Major versions of iOS are released annually.
- The iOS user interface is based upon direct manipulation, using multi-touch gestures. Interface control elements consist of sliders, switches, and buttons.
- Interaction with the OS includes gestures such as swipe, tap, pinch, and reverse pinch, all of which have specific definitions within the context of the iOS operating system and its multi-touch interface.
- **Advantages:-**
 - Provides high customer service
 - Has more security
 - Supports multitasking and performance is awesome
 - Supports vast number of applications
- **Disadvantages**
 - Not flexible, only supports iOS devices
 - Not open source
 - Is costly Apps Developments
 - Not customizable.

Parameter	Android	iOS
Source model	Open source	Closed, with open source components.
OS family	Linux	OS X, UNIX
Language Used	C,c++,JAVA	C++, objective -c
Developer	Google, Open Handset Alliance	Apple Inc.
Widgets	Yes, except on lock screen	No, except in Notification Center
File transfer	Easier than iOS.	More difficult.
Internet browsing	Google Chrome	Safari

Security	Low	High
Performance	Low	Excellent

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